

Third-Order Dual Quaternion (TODQ) Kinematics and Geometrically Consistent Error Systems: A Computed Torque Approach with Certified H_∞ Disturbance Attenuation

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Abstract

Unit dual quaternions provide a globally singularity-free parametrization of rigid-body pose, but their error formulations cover only the pose and velocity levels and therefore do not interface with acceleration-level dynamics. This paper introduces the third-order dual quaternion (TODQ), with which a single product of kinematic elements yields the pose, the twist, and the task-space acceleration of a serial chain. On two hyper-dual quaternion (HDQ) elements we then build an error system that generates a right-invariant pose error and a geometrically consistent twist error in one multiplication, and we show that the resulting error dynamics is a strict cascade. The proposed geometrically consistent computed-torque law endows the closed-loop storage function with an exact dissipation identity, from which nominal convergence, a necessary-and-sufficient condition for L_2 disturbance attenuation, and a mean-square bound for bias-type disturbances follow simultaneously. The L_2 gain can be optimized in closed form: the certified value equals the reciprocal of the smallest damping and coincides with the true H_∞ norm of the linearized loop, so a desired attenuation level converts into a damping lower bound by one division. Thirty-three runs on a six-degree-of-freedom robot arm show that four control laws match each other in tracking accuracy under an equal gain budget, while only the proposed law's attenuation certificate holds throughout, with the certified gain tightened from 1/6 s to 1/15 s.

Keywords: dual quaternion; hyper-dual quaternion; third-order dual quaternion; polynomial algebra; geometrically consistent error; H_∞ control; disturbance attenuation; robot manipulators

1 Introduction

1.1 Motivation

Parametrizing rigid-body pose (orientation plus translation) is the starting point of robot control. The unit dual quaternion packs both into a single eight-dimensional algebraic object whose operations are free of singularities and whose product realizes pose composition directly; on this parametrization, [P2] developed a kinematic tracking controller with an L_2 disturbance-attenuation guarantee, which remains the representative result of dual-quaternion control. Cohen and Shoham [P1] introduced the hyper-dual quaternion (HDQ), which adjoins to a DQ a nilpotent unit ε^* ($\varepsilon^{*2} = 0$) so that a single HDQ element carries a pose together with its first time derivative and chained multiplications differentiate automatically (the Leibniz rule is internalized in the product); forward kinematics then propagates pose and twist in one pass.

HDQ nevertheless falls one order short of the dynamics. Computed-torque control requires the task-space acceleration $\ddot{\mathbf{x}}$ and the Jacobian-derivative term $\dot{J}\dot{\mathbf{q}}$, and in the HDQ structure

of [P1] both nilpotent units are already taken (ε carries translation and ε^* the first derivative), so the $\varepsilon\varepsilon^*$ term carries only the first derivative of the translation part and contains no second-order information. The error system poses a more fundamental obstruction: existing DQ error objects have a pose layer only, and the disturbance model assumes velocity-level additive signals in L_2 . At the dynamics interface three structural gaps therefore appear: the pose error and the velocity error decouple (in laws of the [Ch20] type the pose feedback takes the spiral logarithm, whose derivative map is singular at large errors); acceleration-level disturbances have no entry point; and bias-type uncertainty violates the L_2 assumption, so the available certificates cease to apply.

1.2 Contributions

1. **TODQ kinematics:** based on a three-term polynomial algebra we construct a third-order dual quaternion representation of serial-chain kinematics. It removes the first-derivative ceiling of existing HDQ kinematics: a single $O(n)$ chain product outputs pose, twist, and task-space acceleration simultaneously, supplying the analytic feedforward required by acceleration-level control.
2. **A unified error system:** on two HDQ elements we define an error object that generates a right-invariant pose error and a geometrically consistent twist error in one multiplication, and we prove that the closed-loop error dynamics is a strict cascade.
3. **Control law with exact certificates:** we propose an $A^\top$ -shaping computed-torque law for which the closed-loop storage function satisfies an exact dissipation identity. This yields, in one frame, nominal convergence, a necessary-and-sufficient condition for L_2 attenuation, and a mean-square bound for bias-type disturbances. Optimizing the L_2 gain shows that the certified value equals the reciprocal of the smallest damping and coincides with the true H_∞ norm of the linearized loop, so a desired attenuation level converts into a damping lower bound exactly.
4. **Hardware validation:** thirty-three runs on a six-degree-of-freedom arm—a four-law comparison under an equal gain budget and a γ gain ladder—confirm that every certificate holds on hardware, and separate the transient accuracy governed by the task law from the quasi-static residual dominated by the firmware tether spring and friction.

1.3 Organization

Section 2 fixes notation and recalls quaternions, dual quaternions, and HDQ. Section 3 develops TODQ kinematics. Section 4 constructs the error system and proves the cascade property. Section 5 presents the control law and the three performance certificates. Section 6 reports the hardware experiments, and Section 7 concludes. Complete proofs are collected in the Appendix.

2 Preliminaries

Notation (used throughout): a single pose curve is written with one kernel letter (e.g., x); the ornament above the letter indicates the algebraic layer:

Symbol	Object
$\hat{a}$	quaternion
$\hat{\underline{a}}$	dual quaternion
$\check{a}$	HDQ (hyper-dual quaternion)
$\bar{a}$	TODQ (third-order dual quaternion)
$(\cdot)$	error quantity
boldface	pure DQ (twists, etc.), vectors, and matrices

2.1 Quaternions and dual quaternions

A unit quaternion $\hat{r} = \cos \frac{\phi}{2} + n \sin \frac{\phi}{2}$ represents a rotation by the angle ϕ about the unit axis n .

The dual quaternion (DQ) algebra is $\mathbb{H} \oplus \varepsilon \mathbb{H}$ with $\varepsilon^2 = 0$ (ε commutes with the quaternion units). A **unit DQ**

$$\hat{\underline{x}} = \hat{r} + \varepsilon \frac{1}{2} p \hat{r}, \quad \hat{r} \in \text{Spin}(3), \quad p \in \mathbb{H}_p \text{ (pure quaternion } \cong \mathbb{R}^3) \quad (2.1)$$

represents a pose (rotation $\hat{r}$ plus translation p) and satisfies $\hat{\underline{x}}\hat{\underline{x}}^* = 1$; the unit DQs form the group $\text{Spin}(3) \ltimes \mathbb{R}^3$, which double-covers $SE(3)$, and their product is pose composition. The DQ conjugate $\hat{\underline{x}}^* = \hat{r}^* + \varepsilon \frac{1}{2} \hat{r}^* p^*$ follows the quaternion conjugate componentwise. **Pure DQs** (vanishing scalar and dual-scalar parts) form a six-dimensional real space; vec_6 denotes the coordinate isomorphism extracting the two vector parts, with inverse $\overline{\text{vec}}_6$.

2.2 Twists and kinematics

Along a smooth unit DQ curve $\hat{\underline{x}}(t)$, the **spatial twist** is

$$\boldsymbol{\xi} \triangleq 2\dot{\hat{\underline{x}}}\hat{\underline{x}}^* = \omega + \varepsilon v, \quad v = \dot{p} + p \times \omega, \quad (2.2)$$

with ω the angular velocity. $\boldsymbol{\xi}$ is a pure DQ, and (2.2) is equivalent to the left-multiplied kinematics $\dot{\hat{\underline{x}}} = \frac{1}{2}\boldsymbol{\xi}\hat{\underline{x}}$ (the convention of [P2], Eq. (1)). For an n -joint serial arm, the forward kinematics is $\hat{\underline{x}}(\mathbf{q}) = \prod_{i=1}^n \hat{\underline{x}}_i(q_i)$ (a product of per-joint unit DQs), and differentiation gives $\text{vec}_6 \boldsymbol{\xi} = J(\mathbf{q})\dot{\mathbf{q}}$ with $J \in \mathbb{R}^{6 \times n}$ the geometric Jacobian.

Adjoint action and Lie bracket: $\text{Ad}_{\hat{\underline{x}}} \mathbf{a} \triangleq \hat{\underline{x}} \mathbf{a} \hat{\underline{x}}^*$ transports a twist between frames, and $\text{ad}_{\mathbf{a}} \mathbf{b} \triangleq \frac{1}{2}(\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a})$ is the DQ form of the Lie bracket; both preserve the pure-DQ structure (see [4]).

2.3 Hyper-dual quaternions (HDQ)

An HDQ element has the form $\check{a} = \hat{a}_0 + \varepsilon^* \hat{a}_1$, where both terms $\hat{a}_0, \hat{a}_1$ are DQs (§2.1); ε^* commutes with all quaternion units and $\varepsilon^{*2} = 0$, so the product is fixed by distributivity:

$$(\hat{a}_0 + \varepsilon^* \hat{a}_1)(\hat{b}_0 + \varepsilon^* \hat{b}_1) = \hat{a}_0 \hat{b}_0 + \varepsilon^*(\hat{a}_0 \hat{b}_1 + \hat{a}_1 \hat{b}_0). \quad (2.3)$$

The **HDQ conjugate** (as used in $(\check{x}_d)^*$ of Theorem 1) is the **termwise DQ conjugate**:

$$\check{a}^* \triangleq \hat{a}_0^* + \varepsilon^* \hat{a}_1^*. \quad (2.4)$$

Definition (2.4) makes the conjugate commute with differentiation ($(\check{x})^*$ is the HDQ representation of the curve $\hat{\underline{x}}^*(t)$) and remains an antiautomorphism: $(\check{a}\check{b})^* = \check{b}^* \check{a}^*$.

3 TODQ: Third-Order Dual Quaternions and Kinematic Reconstruction

3.1 The algebra

Definition 1 (TODQ [Con25]) A third-order dual quaternion (TODQ) element has the form

$$\bar{a} = \underline{\hat{a}}_0 + \sigma \underline{\hat{a}}_1 + \frac{1}{2} \sigma^2 \underline{\hat{a}}_2, \quad \underline{\hat{a}}_0, \underline{\hat{a}}_1, \underline{\hat{a}}_2 \text{ all DQs}, \quad (3.1)$$

and the algebra of all TODQ elements is denoted $\mathcal{A}_2$. The unit σ commutes with all quaternion units and satisfies $\sigma^3 = 0$; distributivity together with $\sigma^3 = 0$ fixes the product uniquely:

$$\bar{a} \bar{b} = \underline{\hat{a}}_0 \underline{\hat{b}}_0 + \sigma(\underline{\hat{a}}_0 \underline{\hat{b}}_1 + \underline{\hat{a}}_1 \underline{\hat{b}}_0) + \frac{1}{2} \sigma^2 (\underline{\hat{a}}_0 \underline{\hat{b}}_2 + 2 \underline{\hat{a}}_1 \underline{\hat{b}}_1 + \underline{\hat{a}}_2 \underline{\hat{b}}_0). \quad (3.2)$$

The **TODQ conjugate** is the termwise DQ conjugate (same convention as (2.4)): $\bar{a}^* \triangleq \underline{\hat{a}}_0^* + \sigma \underline{\hat{a}}_1^* + \frac{1}{2} \sigma^2 \underline{\hat{a}}_2^*$, an antiautomorphism of $\mathcal{A}_2$: $(\bar{a} \bar{b})^* = \bar{b}^* \bar{a}^*$.

In other words, $\mathcal{A}_2$ is the truncated polynomial algebra that realizes third-order jets: the nilpotent unit σ indexes the derivative order, so position-, velocity-, and acceleration-level information coexist in one element and separate cleanly by powers of σ .

3.2 Serial-chain kinematics in TODQ form

Given a smooth unit DQ curve $\underline{\hat{x}}(t)$, its **TODQ representation** collects the curve and its first two derivatives into the three coefficients of σ :

$$\bar{x} \triangleq \underline{\hat{x}} + \sigma \dot{\underline{\hat{x}}} + \frac{1}{2} \sigma^2 \ddot{\underline{\hat{x}}} \in \mathcal{A}_2. \quad (3.3a)$$

For a serial arm $\underline{\hat{x}}(\mathbf{q}) = \prod_i \underline{\hat{x}}_i(q_i(t))$, write first the TODQ representation $\bar{x}_i$ of each joint factor (a closed form in $q_i, \dot{q}_i, \ddot{q}_i$ follows from $\partial \underline{\hat{x}}_i / \partial q_i = \frac{1}{2} \mathbf{s}_i \underline{\hat{x}}_i$ and the chain rule), then multiply the factors successively using (3.2):

$$\bar{x} = \bar{x}_1 \bar{x}_2 \cdots \bar{x}_n = \prod_{i=1}^n \bar{x}_i. \quad (3.4)$$

A single $O(n)$ chain product outputs $\underline{\hat{x}}, \dot{\underline{\hat{x}}}, \ddot{\underline{\hat{x}}}$ simultaneously (the three coefficients of $\bar{x}$). Derived quantities:

$$\boldsymbol{\xi} = 2 \dot{\underline{\hat{x}}} \underline{\hat{x}}^*, \quad \dot{\boldsymbol{\xi}} = 2 \ddot{\underline{\hat{x}}} \underline{\hat{x}}^* - \frac{1}{2} \boldsymbol{\xi}^2 \text{ (pure part)}, \quad \text{vec}_6 \dot{\boldsymbol{\xi}} = \dot{J} \dot{\mathbf{q}} + J \ddot{\mathbf{q}}. \quad (3.5)$$

4 Geometrically Consistent Error System

4.1 Which orders the error needs

The feedback part of the computed-torque law (§5) is $-K_d e_\xi - A^\top K_p e_z$ and uses only the pose and velocity levels: the reference acceleration is fed forward (a known quantity determined by the desired trajectory), and acceleration-level uncertainty is assigned to the disturbance $d(t)$ entering the $\dot{e}_\xi$ equation (§5.2). Including an acceleration error would only inject the poorly conditioned differentiated acceleration estimate (variance $\propto \Delta t^{-4}$) into the feedback and add one dimension of dynamics. For this reason, **the forward kinematics and the desired trajectory are modeled in TODQ (the feedforward needs the σ^2 term), while the error system is defined on HDQ.**

4.2 Theorem 1: HDQ representation of the error

The measured chain takes the HDQ representation $\check{x} = \hat{x} + \varepsilon^* \dot{\hat{x}}$, and the desired chain $\check{x}_d = \hat{x}_d + \varepsilon^* \dot{\hat{x}}_d$.

Theorem 1 (HDQ lift of the error) Define the HDQ error element

$$\check{\tilde{x}} \triangleq \check{x} \cdot (\check{x}_d)^*. \quad (4.1)$$

Then

$$\check{\tilde{x}} = \tilde{x} + \varepsilon^* \dot{\tilde{x}}, \quad \tilde{x} = \hat{x} \hat{x}_d^*, \quad \dot{\tilde{x}} = \dot{\hat{x}} \hat{x}_d^* + \hat{x} \dot{\hat{x}}_d^*, \quad (4.2)$$

so a single HDQ multiplication (three DQ multiplications) delivers the pose error $\tilde{x}$ and its derivative simultaneously; the order-zero term is exactly the $\tilde{x}$ of [P2]. Define further the **geometrically consistent twist error**

$$\tilde{\xi} \triangleq 2\dot{\tilde{x}}\tilde{x}^*, \quad e_\xi \triangleq \text{vec}_6 \tilde{\xi}. \quad (4.3)$$

Then: (i) $\tilde{\xi}$ is a pure DQ and is invariant under the unwinding flip $\tilde{x} \rightarrow -\tilde{x}$; (ii) the error satisfies left-multiplied kinematics of the same form as the original system, $\dot{\tilde{x}} = \frac{1}{2}\tilde{\xi}\tilde{x}$; (iii) in the disturbance-free case,

$$\tilde{\xi} = \xi - \text{Ad}_{\tilde{x}} \xi_d, \quad (4.4)$$

that is, the actual twist minus the desired twist **transported to the current pose**.

Proof. For (4.2), apply (2.3) with $\hat{a}_0 = \hat{x}, \hat{a}_1 = \dot{\hat{x}}, \hat{b}_0 = \hat{x}_d^*, \hat{b}_1 = \dot{\hat{x}}_d^*$ (conjugation commuting with differentiation ensures that the ε^* term of $(\check{x}_d)^*$ is $\dot{\hat{x}}_d^*$); the ε^* term is the Leibniz expansion of $\frac{d}{dt}(\hat{x}\hat{x}_d^*)$. (i) follows from $2(-\dot{\tilde{x}})(-\tilde{x}^*) = 2\dot{\tilde{x}}\tilde{x}^*$; (ii) from right-multiplying (4.3) by $\tilde{x}$; (iii): $\dot{\tilde{x}} = \dot{\hat{x}}\hat{x}_d^* + \hat{x}\dot{\hat{x}}_d^* = \frac{1}{2}\xi\tilde{x} - \frac{1}{2}\tilde{x}\xi_d$ (using $\dot{\hat{x}}_d^* = -\frac{1}{2}\hat{x}_d^*\xi_d$), and right-multiplication by $2\tilde{x}^*$ yields (4.4). ■

4.3 Theorem 2: cascade kinematics of the output error

Following [P2], the output is $e_z = [\mathcal{O}; \mathcal{T}] \in \mathbb{R}^6$ with $\mathcal{O} = -\text{Im} \tilde{r}$ and $\mathcal{T} = \tilde{p}$ ($\tilde{r}, \tilde{p}$ the rotation/translation components of $\tilde{x}$, $\tilde{r} = \tilde{\eta} + \tilde{\mu}$).

Theorem 2 (Closed-form output-error kinematics) Writing $\tilde{\xi} = \tilde{\omega} + \varepsilon \tilde{v}$, one has

$$\dot{e}_z = A(\tilde{x}) e_\xi, \quad A(\tilde{x}) = \begin{bmatrix} -\frac{1}{2}(\tilde{\eta}I_3 + [\mathcal{O}]_\times) & 0_3 \\ -[\mathcal{T}]_\times & I_3 \end{bmatrix}, \quad (4.5)$$

and as $\tilde{x} \rightarrow 1$, $A \rightarrow A_0 = \text{diag}(-\frac{1}{2}I_3, I_3)$ with $\sigma_{\min}(A_0) = \frac{1}{2}$.

Proof. Rotation: by $\dot{\tilde{r}} = \frac{1}{2}\tilde{\omega}\tilde{r}$ (quaternion part of Theorem 1(ii)), $\dot{\mathcal{O}} = -\text{Im}(\frac{1}{2}\tilde{\omega}\tilde{r}) = -\frac{1}{2}(\tilde{\eta}I_3 + [\mathcal{O}]_\times)\tilde{\omega}$ (using $\tilde{\omega} \times \tilde{\mu} = [\mathcal{O}]_\times\tilde{\omega}$). Translation: the twist convention $\tilde{v} = \dot{\tilde{p}} + \tilde{p} \times \tilde{\omega}$ gives $\dot{\mathcal{T}} = \tilde{v} - [\mathcal{T}]_\times\tilde{\omega}$. ■

Theorems 1 and 2 give a strict two-layer cascade: $e_z \xrightarrow{A} e_\xi$ with $\dot{e}_z = A e_\xi$, while $\dot{e}_\xi$ is governed by the control law and the disturbance (§5). The error state is 12-dimensional and contains no acceleration layer—the structural choice argued in §4.1.

5 Geometrically Consistent Control Law and Certified H_∞ Performance

5.1 Disturbances: explicit solution, multiplicative/exogenous separation, assumptions

The true joint-space dynamics with friction $\mathbf{f}$, actuation error $\delta\boldsymbol{\tau}$, and environmental contact $\boldsymbol{\tau}_{\text{ext}}$ is $M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \mathbf{f} = \boldsymbol{\tau} + \delta\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{ext}}$; the inner loop executes the computed torque based on the nominal model, $\boldsymbol{\tau} = \hat{M}\ddot{\mathbf{q}}_{\text{ref}} + \hat{C}\dot{\mathbf{q}} + \hat{\mathbf{g}} + \hat{\mathbf{f}}$. With the mismatch defined as “nominal minus true”, $\Delta M \triangleq \hat{M} - M$ (likewise $\Delta C, \Delta \mathbf{g}, \Delta \mathbf{f}$), substituting and left-multiplying by M^{-1} yields $\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_{\text{ref}} + \mathbf{w}_{\text{dyn}}$ with

$$\mathbf{w}_{\text{dyn}} = M^{-1}(\Delta M \ddot{\mathbf{q}}_{\text{ref}} + \Delta C \dot{\mathbf{q}} + \Delta \mathbf{g} + \Delta \mathbf{f} + \delta\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{ext}}). \quad (5.1)$$

Separation into multiplicative and exogenous parts. Substituting $\ddot{\mathbf{q}}_{\text{ref}} = J^+(u_{\text{ff}} + u_{\text{fb}} - \dot{J}\dot{\mathbf{q}})$ into (5.1) and using $JJ^+ = I_6$ gives the disturbance decomposition

$$d = \Theta(\mathbf{q}) u_{\text{fb}} + d_{\text{ex}}, \quad \Theta \triangleq JM^{-1}\Delta M J^+ \in \mathbb{R}^{6 \times 6}, \quad (5.1d)$$

where Θ is the multiplicative part and d_{ex} the exogenous part (see Appendix C.1). The exogenous part is further split by time profile, $d_{\text{ex}} = d_{L_2} + d_b$: the former (L_2 , noise-like) is handled by Theorem 3(c) and the latter (L_∞ , bias-like) by Theorem 3(d).

Assumptions (used silently by Theorem 3, listed here once):

- **(A1) Regular Jacobian:** on a working set $\mathcal{Q}$, $J(\mathbf{q})$ has full row rank with $\sigma_{\min}(J) \geq \underline{\sigma} > 0$, hence $JJ^+ = I_6$ (near singularities a damped pseudoinverse is used [Nak86], and the residual is folded into d_{ex}).
- **(A2) Bounded inertia and mismatch:** $M(\mathbf{q}) \succeq \underline{m}I_n$ ($\underline{m} > 0$), and $\Delta M, \Delta C, \Delta \mathbf{g}, \Delta \mathbf{f}$ are bounded on $\mathcal{Q} \times \{\|\dot{\mathbf{q}}\| \leq \bar{v}\}$.
- **(A3) Multiplicative small-gain condition** [ZDG96]:

$$\alpha \triangleq \sup_{\mathbf{q} \in \mathcal{Q}} \|\Theta(\mathbf{q})\|_2 < \frac{\lambda_{\min}(K_d)}{\lambda_{\max}(K_d)} = \frac{1}{\text{cond}_2(K_d)} \leq 1. \quad (5.1f)$$

For isotropic damping this reduces to the classical robustness condition $\alpha < 1$ of computed-torque control [Spo92].

- **(A4) Desired trajectory and exogenous signals:** $\hat{\mathbf{x}}_d(t)$ is C^2 with bounded $\|\dot{\boldsymbol{\xi}}_d\|, \|\ddot{\boldsymbol{\xi}}_d\|$; $d_{L_2} \in L_2$, $d_b \in L_\infty$, and $D_{\text{ex}} \triangleq \|d_{\text{ex}}\|_{L_\infty} < \infty$ (boundedness of $\dot{\mathbf{v}}_w$ requires a filtered derivative in implementation [Ber93]).

5.2 Control law

Take K_p symmetric positive definite (typically block diagonal, $K_p = \text{diag}(p_O I_3, p_T I_3)$; $K_p = k_p I_6$ recovers the scalar case), and K_d symmetric positive definite, block diagonal $K_d = \text{diag}(K_\omega, K_v)$ when the per-channel bounds of Theorem 3(c-2) are desired. The reference acceleration (the geometrically consistent computed-torque law [LWP80]) is

$$\ddot{\mathbf{q}}_{\text{ref}} = J^+ \left(\underbrace{\text{vec}_6(\text{Ad}_{\tilde{\mathbf{x}}} \dot{\boldsymbol{\xi}}_d + \text{ad}_{\tilde{\boldsymbol{\xi}}} \text{Ad}_{\tilde{\mathbf{x}}} \boldsymbol{\xi}_d)}_{\text{feedforward: transported desired acceleration} + \text{transport correction}} - K_d e_\xi - A^\top(\tilde{\mathbf{x}}) K_p e_z - \dot{J}\dot{\mathbf{q}} \right). \quad (5.2)$$

The pose feedback is $A^\top$ -shaped: taking $-A^\top(\tilde{\mathbf{x}})K_p e_z$ rather than, e.g., $-K_p \log\text{-errors}$, makes the cross term in the Lyapunov derivative cancel exactly (the role of $A^\top$ is that of an energy-shaping potential expressed in the error coordinates). The feedforward and $\dot{J}\dot{\mathbf{q}}$ come from the σ^2 term of the desired chain and from (3.5); all feedback quantities are read off the HDQ error element of Theorem 1, and no acceleration error is measured or estimated.

5.3 Main results

The four results of this section (Theorems 3(a)–3(d)) share assumptions (A1)–(A4) (§5.1), the control law (5.2), and the disturbance model (5.1)/(5.1d); the common storage function is

$$V(e_z, e_\xi) = \frac{1}{2} \|e_\xi\|^2 + \frac{1}{2} e_z^\top K_p e_z \geq 0, \quad (5.4a)$$

with $K_p \succ 0$ symmetric ($K_p = k_p I_6$ recovers the scalar form $\frac{k_p}{2} \|e_z\|^2$). Limited by the topology of $SO(3)$ (double cover) and the singularity of A at $\tilde{\eta} = 0$, all convergence claims are local, on a sublevel set.

Theorem 3(a) (Closed-loop error dynamics) The error coordinates (e_z, e_ξ) satisfy the cascade

$$\dot{e}_z = A(\tilde{x}) e_\xi, \quad \dot{e}_\xi = -K_d e_\xi - A^\top(\tilde{x}) K_p e_z + d(t), \quad (5.5)$$

where $d = J\mathbf{w}_{\text{dyn}} + \dot{\mathbf{v}}_w + \dot{\mathbf{v}}_c$ collects all acceleration-level disturbances and decomposes exactly as $d = \Theta u_{\text{fb}} + d_{\text{ex}}$ by (5.1d).

Proof. Differentiating (4.4) term by term and using the adjoint transport rule $\frac{d}{dt}(\text{Ad}_{\tilde{x}} \mathbf{a}) = \text{Ad}_{\tilde{x}} \dot{\mathbf{a}} + \text{ad}_{\tilde{\xi}}(\text{Ad}_{\tilde{x}} \mathbf{a})$ (direct expansion from $\dot{\tilde{x}} = \frac{1}{2} \tilde{\xi} \tilde{x}$ and its conjugate) gives the twist-error dynamics

$$\dot{\tilde{\xi}} = \dot{\xi} - \text{Ad}_{\tilde{x}} \dot{\xi}_d - \text{ad}_{\tilde{\xi}}(\text{Ad}_{\tilde{x}} \xi_d) \quad (5.5a)$$

(with disturbances, $\dot{\mathbf{v}}_w + \dot{\mathbf{v}}_c$ is added on the right). Substituting into this the $\dot{\tilde{\xi}}$ obtained from (5.2) via (3.5), the feedforward cancels the desired/transport terms exactly and (5.5) results. ■

Theorem 3(b) (Nominal case: sublevel-set invariance, asymptotic and local exponential convergence) Let $d \equiv 0$, and let $K_p = \text{diag}(K_{p,O}, K_{p,T})$ be symmetric positive definite (rotation/translation blocks). Take

$$0 < c < c^* \triangleq \frac{1}{2} \lambda_{\min}(K_{p,O}), \quad \Omega_c \triangleq \{(e_z, e_\xi) : V \leq c\}. \quad (5.5b)$$

Then: (i) $\dot{V} = -e_\xi^\top K_d e_\xi \leq 0$, so Ω_c is compact and forward invariant; (ii) if $\tilde{\eta}(0) > 0$ then along the trajectory

$$\tilde{\eta}(t) \geq \eta_0 \triangleq \sqrt{1 - \frac{2c}{\lambda_{\min}(K_{p,O})}} > 0, \quad (5.5c)$$

so $A(\tilde{x})$ is uniformly invertible on Ω_c ($\det A = -\frac{1}{8} \tilde{\eta} \leq -\frac{1}{8} \eta_0 < 0$); (iii) every trajectory in Ω_c satisfies $(e_z, e_\xi) \rightarrow (0, 0)$; (iv) $(0, 0)$ is **locally exponentially stable**.

Proof. The key step is the exact cancellation of the cross term $e_z^\top K_p A e_\xi$ by the $A^\top$ -shaped feedback, giving $\dot{V} = -e_\xi^\top K_d e_\xi$; LaSalle's theorem together with uniform invertibility of A on Ω_c (Appendix C.2) yields asymptotic convergence. See Appendix C.6 for the full proof. ■

Remark 1 The sublevel-set condition (5.5b) is numerically verifiable: for the hardware tier G4 below, $c^* = 72$ while the measured peak of V is about 0.5, a margin of more than two orders of magnitude (§6.1). On the $\tilde{\eta} < 0$ branch, flipping the sign of the HDQ error (which leaves $\tilde{\xi}$ unchanged, Theorem 1(i)) enforces $\tilde{\eta} \geq 0$. ■

Theorem 3(c) (L_2 disturbances: H_∞ quadratic/Schur condition and per-channel split) Let $d = d_{L_2} \in L_2$.

Certificate parameters: $\kappa > 0$ is the reciprocal of the error penalty (the supply-rate coefficient of $\|e_\xi\|^2$ is $\frac{1}{2\kappa}$), and $\gamma_a > 0$ is the certified attenuation level; the certified quantity is the L_2 gain $\gamma^* \triangleq \gamma_a \sqrt{\kappa}$ from d to e_ξ (in seconds: d is an acceleration-level and e_ξ a velocity-level

signal, and for zero initial state $\|e_\xi\|_{L_2} \leq \gamma^* \|d_{L_2}\|_{L_2}$). Both are analysis parameters: they never enter the control law (5.2), but by Theorem 3(c') they invert into a tuning rule—certifying a target gain γ^* is equivalent to the damping lower bound $K_d \succeq (1/\gamma^*)I_6$. In the per-channel version, $(\kappa_\omega, \gamma_\omega)$ and (κ_v, γ_v) can be specified independently.

(c-1) Pooled condition (arbitrary symmetric positive definite K_d): the performance target $\dot{V} \leq -\frac{1}{2\kappa}\|e_\xi\|^2 + \frac{\gamma_a^2}{2}\|d\|^2$ holds for all (e_ξ, d) **if and only if**

$$M \triangleq \begin{bmatrix} K_d - \frac{1}{2\kappa}I_6 & -\frac{1}{2}I_6 \\ -\frac{1}{2}I_6 & \frac{\gamma_a^2}{2}I_6 \end{bmatrix} \succeq 0 \xLeftrightarrow{\text{Schur}} K_d \succeq \frac{1}{2}(\kappa^{-1} + \gamma_a^{-2})I_6, \quad (5.6a)$$

in which case

$$\int_0^\infty \kappa^{-1}\|e_\xi\|^2 dt \leq \gamma_a^2 \int_0^\infty \|d_{L_2}\|^2 dt + 2V(0), \quad (5.6)$$

i.e., the L_2 gain from the acceleration-level disturbance to the twist-error energy is at most $\gamma_a\sqrt{\kappa}$ (for zero initial condition this reduces to a pure gain bound; the bound constrains e_ξ only).

(c-2) Per-channel condition: let $K_d = \text{diag}(K_\omega, K_v)$ be block diagonal and let $K_p = \text{diag}(K_{p,O}, k_{p,T}I_3)$ with the translational block isotropic (necessity in Appendix C.3). Writing $e_\xi = [\tilde{\omega}; \tilde{v}]$, $d = [d_\omega; d_v]$, and $V_\omega \triangleq \frac{1}{2}\|\tilde{\omega}\|^2 + \frac{1}{2}\mathcal{O}^\top K_{p,O}\mathcal{O}$, $V_v \triangleq \frac{1}{2}\|\tilde{v}\|^2 + \frac{k_{p,T}}{2}\|\mathcal{T}\|^2$ ($V = V_\omega + V_v$), if

$$K_\omega \succeq \frac{1}{2}(\kappa_\omega^{-1} + \gamma_\omega^{-2})I_3, \quad K_v \succeq \frac{1}{2}(\kappa_v^{-1} + \gamma_v^{-2})I_3, \quad (5.6b)$$

then the rotation and translation channels satisfy independently

$$\int_0^\infty \kappa_\omega^{-1}\|\tilde{\omega}\|^2 dt \leq \gamma_\omega^2 \int_0^\infty \|d_\omega\|^2 dt + 2V_\omega(0), \quad \int_0^\infty \kappa_v^{-1}\|\tilde{v}\|^2 dt \leq \gamma_v^2 \int_0^\infty \|d_v\|^2 dt + 2V_v(0), \quad (5.6')$$

with dimensionally homogeneous quantities in each channel, and $(\kappa_\omega, \gamma_\omega)$, (κ_v, γ_v) specifiable independently.

Proof. With disturbances the cross terms still cancel exactly, giving $\dot{V} = -e_\xi^\top K_d e_\xi + e_\xi^\top d$; writing the performance target as $[e_\xi; d]^\top M [e_\xi; d] \geq 0$ and taking the Schur complement yields the necessary-and-sufficient condition (5.6a); the per-channel decoupling rests on two algebraic identities ($\mathcal{T} \times \mathcal{T} = 0$ and $\mathcal{T} \cdot (\mathcal{T} \times \tilde{\omega}) = 0$). See Appendix C.7. ■

Theorem 3(c') (Optimal certificate and inversion of the target gain γ^*) Let $\gamma^* \triangleq \gamma_a\sqrt{\kappa}$ denote the certified L_2 gain and $\lambda \triangleq \lambda_{\min}(K_d) > 0$. Then: (i) (lower bound) every feasible point of the certificate family (5.6a) satisfies $\gamma^* \geq 1/\lambda$; (ii) (attainability) $(\kappa, \gamma_a) = (\lambda^{-1}, \lambda^{-1/2})$ is feasible and gives $\gamma^* = 1/\lambda$. Hence the optimal (smallest) gain certifiable by the family is

$$\gamma_{\text{opt}}^* = \min_{(\kappa, \gamma_a) \text{ feasible}} \gamma_a\sqrt{\kappa} = \frac{1}{\lambda_{\min}(K_d)}, \quad (5.6c)$$

attained at $\kappa = \gamma_a^2 = 1/\lambda_{\min}(K_d)$, where (5.6a) holds with equality. Equivalently, the **inversion rule**: a target gain γ^* is certifiable **if and only if** $K_d \succeq (1/\gamma^*)I_6$, and the leanest certificate is $(\kappa, \gamma_a) = (\gamma^{*2}, \sqrt{\gamma^*})$. The per-channel version (5.6b) likewise gives $\gamma_\omega^* = 1/\lambda_{\min}(K_\omega)$ and $\gamma_v^* = 1/\lambda_{\min}(K_v)$.

Proof. (i) By the AM–GM inequality (all terms positive), $\kappa^{-1} + \gamma_a^{-2} \geq 2\sqrt{\kappa^{-1}\gamma_a^{-2}} = \frac{2}{\gamma_a\sqrt{\kappa}} = \frac{2}{\gamma^*}$; substituting into (5.6a) gives $\lambda \geq \frac{1}{2}(\kappa^{-1} + \gamma_a^{-2}) \geq 1/\gamma^*$. (ii) At $\kappa = \gamma_a^2 = \lambda^{-1}$ we have $\kappa^{-1} + \gamma_a^{-2} = 2\lambda$, so (5.6a) holds with equality and the certified gain is $\gamma_a\sqrt{\kappa} = 1/\lambda$; moreover $\kappa = \lambda^{-1} > 1/(2\lambda)$, so the feasible set is nonempty. Necessity of the inversion rule is (i); sufficiency: if $\lambda \geq 1/\gamma^*$, the point $(\kappa, \gamma_a) = (\gamma^{*2}, \sqrt{\gamma^*})$ satisfies $\kappa^{-1} + \gamma_a^{-2} = 2/\gamma^* \leq 2\lambda$, is feasible, and certifies exactly $\gamma_a\sqrt{\kappa} = \gamma^*$. ■

Remark 2 Theorem 3(c') collapses the pair (κ, γ_a) into a single conversion: the optimal certified gain equals $1/\lambda_{\min}(K_d)$. The tuning entry is the damping: a target attenuation level inverts into a lower bound on K_d ; under the pole-placement parametrization of §5.4, $\gamma^* = 1/(a + b)$. This is equivalent to writing γ directly into the law as in first-order H_∞ schemes, except that here the knob is the physical gain K_d . ■

Theorem 3(d) (L_∞ disturbances: multiplicative separation and mean-square bound on the twist error) Let (A1)–(A4) hold, let the disturbance decompose as in (5.1d), $d = \Theta u_{\text{fb}} + d_{\text{ex}}$, with $\alpha \triangleq \sup_t \|\Theta\|_2$ satisfying the small-gain condition (5.1f), and let $D_{\text{ex}} \triangleq \|d_{\text{ex}}\|_{L_\infty} < \infty$. Define the **effective damping**

$$\lambda_{\text{eff}} \triangleq \lambda_{\min}(K_d) - \alpha \lambda_{\max}(K_d) > 0 \quad (5.7a)$$

(positive by (5.1f)). Suppose the trajectory remains in Ω_c of (5.5b) over the interval considered (this premise only bounds $\|e_z\|$ and is verified a posteriori; see the remark below), and define the **equivalent disturbance amplitude**

$$D \triangleq D_{\text{ex}} + \alpha \lambda_{\max}(K_p) \left(1 + \sqrt{\frac{2c}{\lambda_{\min}(K_{p,T})}}\right) \sqrt{\frac{2c}{\lambda_{\min}(K_p)}}. \quad (5.7b)$$

Then the twist error satisfies the finite-time mean-square bound

$$\frac{1}{T} \int_0^T \|e_\xi(t)\|^2 dt \leq \frac{D^2}{\lambda_{\text{eff}}^2} + \frac{2V(0)}{\lambda_{\text{eff}} T}, \quad \forall T > 0, \quad (5.7c)$$

hence

$$\limsup_{T \rightarrow \infty} \text{RMS}_{[0,T]}(e_\xi) \triangleq \limsup_{T \rightarrow \infty} \left(\frac{1}{T} \int_0^T \|e_\xi\|^2 dt \right)^{1/2} \leq \frac{D}{\lambda_{\text{eff}}} \xrightarrow{\alpha \rightarrow 0} \frac{\|d_{\text{ex}}\|_{L_\infty}}{\lambda_{\min}(K_d)}. \quad (5.7)$$

That is, bias-type (L_∞) uncertainty does not destroy boundedness; it only raises the steady mean-square level of the twist error to D/λ_{eff} . The multiplicative term Θu_{fb} acts twice—it reduces the effective damping (denominator) by $\alpha \lambda_{\max}(K_d)$ and raises the equivalent disturbance amplitude (numerator) by $\alpha \lambda_{\max}(K_p) \|e_z\|$ —and both effects vanish as $\alpha \rightarrow 0$. The pose error e_z is not covered by this theorem; its steady order of magnitude is estimated quasi-statically by the near-identity model (5.9).

Proof. Split Θu_{fb} into a damping-reduction term and an equivalent disturbance term, absorb each with Young's inequality, and integrate to obtain the RMS bound (5.7). See Appendix C.8. ■

Remark 3 Since the negative term of $\dot{V}$ contains only $-\|e_\xi\|^2$ (the disturbance has relative degree two with respect to e_z), the result is an RMS bound rather than a pointwise ISS bound; strictification is required for the latter (Appendix C.4). The Ω_c premise must be verified a posteriori (§6.1). ■

5.4 Near-identity linearization and the static stiffness scaling law

Theorem 3 is qualitative and energetic; it does not return gain values. Linearizing (5.5) around the origin yields a two-channel second-order model directly usable for gain tuning and experimentally falsifiable.

With $K_d = \text{diag}(K_\omega, K_v)$ and $K_p = \text{diag}(K_{p,O}, k_{p,T} I_3)$, as $\tilde{x} \rightarrow 1$ we have $A \rightarrow A_0 = \text{diag}(-\frac{1}{2} I_3, I_3)$, and eliminating e_ξ gives the two decoupled second-order equations

$$\ddot{O} + K_\omega \dot{O} + \frac{1}{4} K_{p,O} O = -\frac{1}{2} d_\omega, \quad \ddot{T} + K_v \dot{T} + k_{p,T} T = +d_v. \quad (5.8)$$

The rotation block $-\frac{1}{2}I_3$ makes the effective rotational stiffness $\frac{1}{4}K_{p,O}$; to match bandwidth across channels take $K_{p,O} = 4k_{p,T}I_3$. Setting $(\ddot{\cdot}) = (\dot{\cdot}) = 0$ yields the **static stiffness scaling law**

$$\|\mathcal{T}\|_{ss} = \frac{\|d_v\|}{k_{p,T}}, \quad \|\mathcal{O}\|_{ss} = \frac{2\|d_\omega\|}{\lambda(K_{p,O})} : \quad (5.9)$$

the steady residual is governed *only* by the static stiffness, not by the damping. Pole-placement rule: for target poles $\{-a, -b\}$ per channel ($a, b > 0$),

$$K_\omega = K_v = (a + b)I_3, \quad k_{p,T} = ab, \quad K_{p,O} = 4ab I_3.$$

Combined with Theorem 3(c'), the optimal certified gain under this parametrization is

$$\gamma^* = \frac{1}{a + b} : \quad \text{the tuning entry of } \gamma \text{ is the pole sum } a + b \text{ (damping),}$$

while the steady residual is set independently by $k = ab$ (stiffness), via (5.9)—each performance metric has a single dedicated knob.

Two experimentally falsifiable predictions follow from (5.9): **(P1)** multiplying the stiffness by ρ divides the steady residual by ρ ; **(P2)** the disturbance levels ($\|d_v\|, \|d_\omega\|$) inverted from different gain tiers under the same physical conditions should coincide. (5.8)–(5.9) are local approximations as $\tilde{x} \rightarrow 1$; the rigorous statements are those of Theorem 3. The verification premise of (P2)—that the task stiffness is the dominant spring—is not met under the hardware tether protocol of §6.2, so its tether-free verification is left as future work.

Tightness (Appendix C.9): the certified value $1/\lambda_{\min}(K_d)$ equals the true H_∞ gain of the closed-loop linearization, so the certificate is free of conservatism and the worst-case disturbance frequency is the channel natural frequency $\sqrt{c}$ (6 rad/s for hardware tier G4).

6 Hardware Validation

6.1 Platform, protocol, and experimental matrix

The platform is a six-degree-of-freedom B601-DM manipulator: joint bus and state feedback at 500 Hz, control period $\Delta t = 10$ ms, and a torque interface $\tau = \hat{M}\ddot{\mathbf{q}}_{\text{ref}} + \hat{C}\dot{\mathbf{q}} + \hat{\mathbf{g}}$ augmented by feedforward model-completion terms (viscous friction 0.8, Coulomb friction $[0, 0.08, 1.69, 0, 0, 0]$ N m, and rotor inertia $\text{diag}(0.01)$; none of these are feedback). The task is a 10 s point-to-point min-jerk motion to a fixed target pose ($\max \|\dot{\mathbf{q}}\| = 0.253$ rad/s, $\max \|\ddot{\mathbf{q}}\| = 2.44$ rad/s²). The experimental matrix combines four control laws with DC stiffness levels $k \in \{8, 16, 24, 36\}$ s⁻² and firmware tether springs $K_{tr} \in \{12, 30, 60\}$ N m/rad, where the four laws are: C1 (proposed, (5.2)); C2, the spiral-logarithm second-order baseline of [Ch20]; C3, the first-order H_∞ law of [P2] bridged to acceleration level through a velocity servo with $k_{\text{servo}} = 20$; and C4, a classical decoupled SE(3) second-order law with quaternion-vector pose error. The tether spring pulls along the planned reference trajectory (not to a fixed anchor) and is lowered by one order of magnitude from the vendor default of 120/18 in order to purify the task law. The $k=36$ tier was run 22 times (one run per setting enters the figures and tables), and all 33 valid runs produced zero torque saturation, with the acceleration governor active in at most 11.2% of frames.

All laws are balanced according to the transmission factors of the linearization (5.8) so that both channels share the same DC stiffness k (C1: $p_O=4k$, $p_T=k$; C2: $K_P=kI_6$; C4: $K_{p,O}=2k$, $K_{p,T}=k$; C3: $k_O=\sqrt{2}/\gamma_O$, $k_T=\sqrt{2}/\gamma_T$ with servo bridging), and the second-order laws take $K_d = a + b$, $k = ab$ (poles $\{-a, -b\}$ in both channels). By Theorem 3(c'), the certified L_2 gain $d \rightarrow e_\xi$ takes its optimal value $\gamma^* = 1/K_d = 1/(a + b)$, with optimal certificate point $(\kappa, \gamma_a) = (\gamma^*, \sqrt{\gamma^*})$; the gain ladder of Table 1 is thus a γ ladder, tightening γ^* from 1/6 to 1/15 s. Two

Table 1: Hardware gain ladder and certified γ ladder (matched DC stiffness $k = ab$; $\gamma^* = 1/(a+b)$ by Theorem 3(c'), with certificate point $(\kappa, \gamma_a) = (\gamma^*, \sqrt{\gamma^*})$). The bracketed (γ_O, γ_T) of C3 are that law's internal H_∞ parameters ($k_O = \sqrt{2}/\gamma_O$, $k_T = \sqrt{2}/\gamma_T$, $k_{\text{servo}} = 20$), unrelated to the certified gain γ^* . [†]At the discretization boundary ($b \Delta t = 0.15$), not run on hardware.

[‡]Reference configuration for 5 ms control; violates the discretization constraint at 100 Hz.

Tier	(a, b)	C1 K_d ; (p_O, p_T)	C2 K_v ; K_P	C3 (γ_O, γ_T)	C4 K_d ; $(K_{p,O}, K_{p,T})$	certified γ^* (s)
G2	(2, 4)	–	6; $8I_6$	(1.768, 3.536)	–	1/6
G3	(2, 8)	10; (64, 16)	10; $16I_6$	(0.884, 1.768)	10; (32, 16)	1/10
G3.5	(3, 8)	11; (96, 24)	11; $24I_6$	(0.589, 1.179)	11; (48, 24)	1/11
G4	(3, 12)	15; (144, 36)	15; $36I_6$	(0.393, 0.786)	15; (72, 36)	1/15
G5 [†]	(3, 15)	18; (180, 45)	18; $45I_6$	(0.314, 0.629)	18; (90, 45)	1/18
G6 [‡]	(4, 20)	24; (320, 80)	24; $80I_6$	(0.177, 0.354)	24; (160, 80)	1/24

hardware constraints bound the ladder: the discretization constraint $\max |\text{pole}| \cdot \Delta t \leq 0.15$ gives $b \leq 15$ at $\Delta t = 10$ ms (tier $k=36$ uses $b=12$, a 20% margin; the reference tier G6 with $b=20$, $\gamma^* = 1/24$ s, corresponds to a 5 ms design and is infeasible at 100 Hz), so the smallest certifiable γ^* on this arm is 1/18 s (boundary tier G5); and the acceleration budget $0.53 K_d + 0.05 k \leq 12 \text{ rad/s}^2$ (proxy peak ≈ 9.8 at $k=36$). Isotropic K_d gives $\text{cond}_2(K_d) = 1$, the widest budget $\alpha < 1$ for (5.1f); and the sublevel constant $c^* = \frac{1}{2} \lambda_{\min}(K_{p,O}) = 72$ exceeds the measured peak of V (≈ 0.5 , worst run) by more than two orders of magnitude.

6.2 Results and analysis

Gain ladder: transients improve monotonically with k . Fig. 1 (tether 30 N m/rad) shows that the p95 orientation error decreases monotonically with k for all four laws (by 14%–20% from $k=8$ to 36, and by 29% for C1 from $k=16$), while the p95 position error decreases mildly (4%–19%)—consistent with the tightening of γ^* from 1/6 to 1/15 s, i.e., with increased damping K_d (Theorem 3(c')).

The quasi-static residual is governed by the tether spring and friction. Unlike an ideal tether-free setup, the hardware steady-state residual is insensitive to the task stiffness k (changes $\leq 10\%$ from $k=16$ to 36) and varies nearly inversely with the tether stiffness: $K_{\text{tr}} = 60/30/12$ N m/rad yields steady position residuals of about 5.5/17/30 mm (Fig. 2). Mechanism: near zero speed the friction feedforward fades out, and the residual is set by the balance between joint friction and the softest spring (the tether), so the task stiffness is not the dominant spring—the inversion premise of the scaling law (5.9) fails under this protocol, and its tether-free quantitative verification is left as future work.

Four laws at equal budget: identical accuracy, difference in certificates. Fig. 3 ($k=36$, tether 30, single runs) shows that within one tether level the four laws differ in steady and p95 errors by at most 2.4 mm and 0.35° —the same order as the run-to-run spread (0.7–9.6 mm). The coincidence of accuracy itself identifies where the contribution of C1 lies: not in tracking accuracy but in the **certificates verifiable throughout**. The other three laws carry no Theorem 3-type certificate (the pose feedback of C2/C4 is not $A^\top$ -shaped and C3 is a first-order bridge), whereas for the proposed law C1 all 10 runs simultaneously satisfy the certificate chain: gains inverted from the target $\gamma^* = 1/K_d$ via Theorem 3(c') with (5.6a) at equality (the optimal certificate point), $\text{cond}_2(K_d) = 1$ with the widest budget $\alpha < 1$ of (5.1f) thanks to isotropic K_d , and a sublevel margin above two orders of magnitude ((5.5b))—without relying on the tether spring to carry the tracking stiffness (K_{tr} can be lowered to 12 N m/rad while the loop remains stable; only the residual rises, by the mechanism above).

Limitations. The validation uses a single target pose and no exogenous disturbance injection; the quasi-static residual is dominated by the firmware tether spring and joint friction, and quantitative verification of (5.9) on hardware requires a tether-free setup or a task stiffness

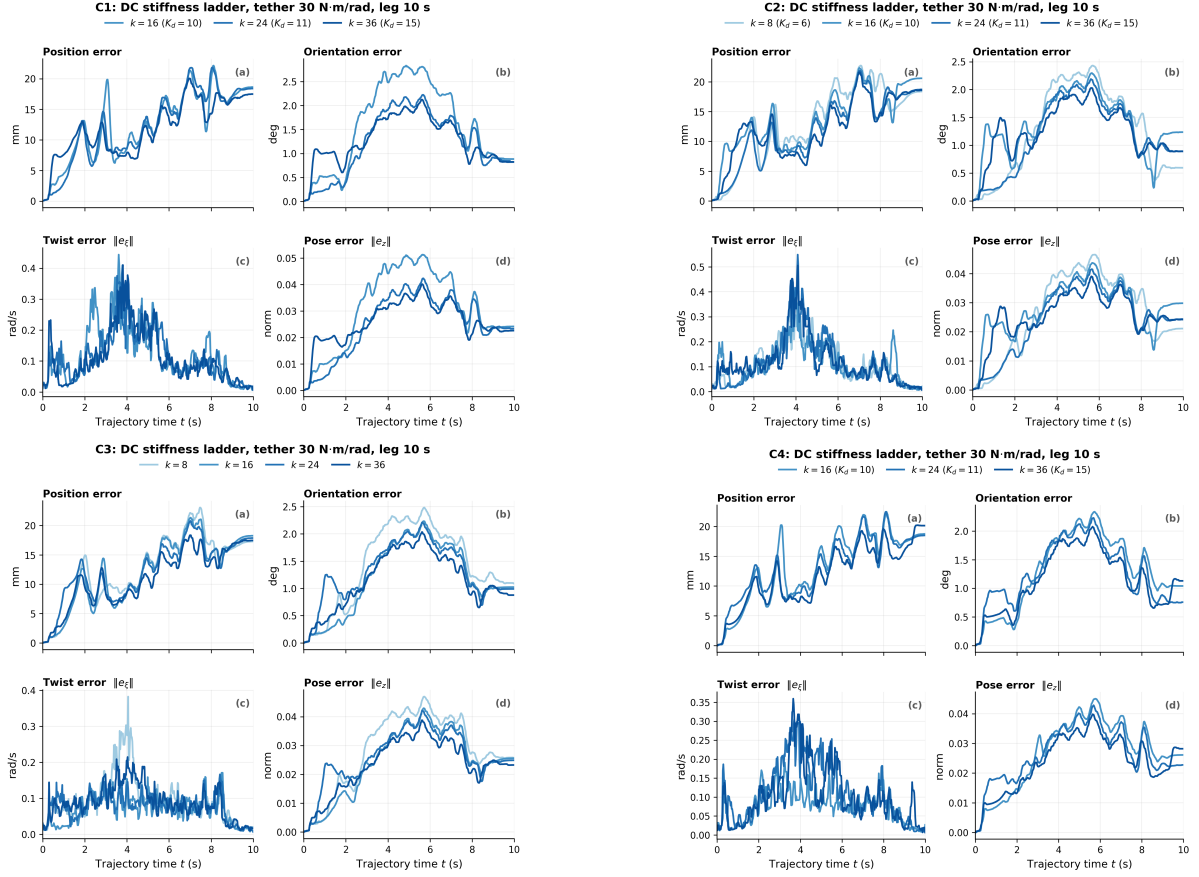


Figure 1: Gain ladder per law (tether 30 N-m/rad, the tiers of $k \in \{8, 16, 24, 36\}$ available to each law, single runs of the 10s nominal trajectory, aligned on traj_t): the orientation and twist transients decrease monotonically with k and the position error mildly so, in the direction of tightening γ^* (Theorem 3(c')).

well above the tether stiffness (left as future work); quantitative large-initial-error experiments are also left as future work.

7 Conclusion

This paper starts from an observation: dynamics-level control requires an acceleration-level task-space model, whereas dual-quaternion formulations typically stop at the velocity level. Filling in that order with third-order dual quaternions makes pose, velocity, and task-space acceleration available from a single kinematic product, lets a single multiplication generate the pose error and the twist error together, and removes the need for a separate acceleration-level error. The resulting structure endows the closed-loop storage function with an exact dissipation identity, so that nominal convergence, L_2 disturbance attenuation, and the steady bound for bias-type disturbances are proved in one framework and map onto one set of physical gains. Most valuable for tuning is the exactness of the attenuation metric: the certified L_2 gain equals the reciprocal of the smallest damping and coincides with the true H_∞ norm of the linearized loop, so inferring a damping lower bound from a disturbance level is one division, not an estimate. The near-identity linearization also yields a practical warning—the rotational stiffness suffers a factor-1/4 reduction, which the hardware gain configuration compensates.

The hardware results support this reading. Under an equal gain budget the proposed law matches three representative baselines in tracking accuracy; the difference lies in provability:

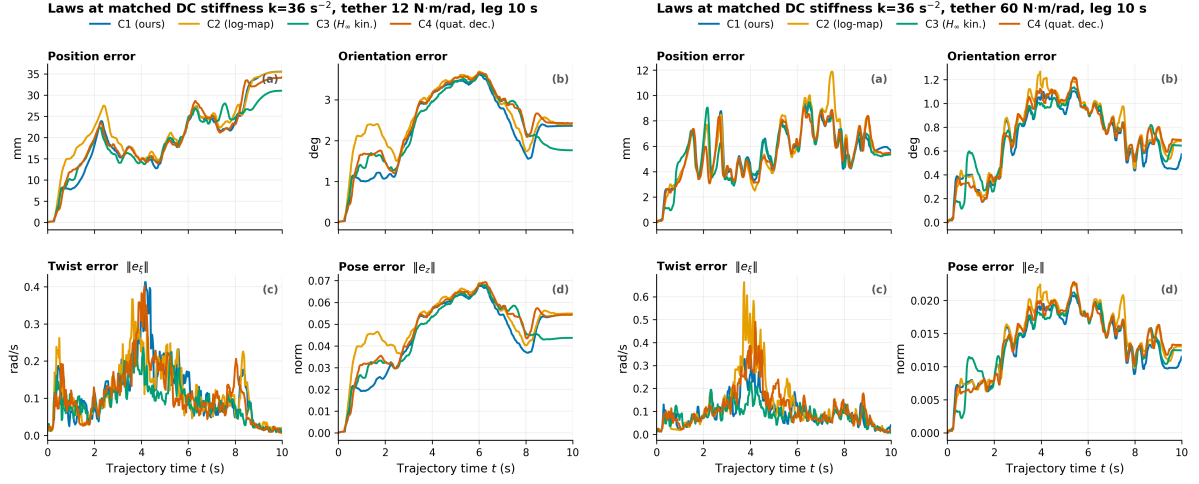


Figure 2: Tether sweep with four laws ($k=36$): left, $K_{tr}=12$ N·m/rad (friction-balance dominated, steady residual ≈ 30 mm); right, $K_{tr}=60$ N·m/rad (millimeter level, steady ≈ 5.5 mm). The residual varies nearly inversely with the tether stiffness while the four laws remain on top of each other.

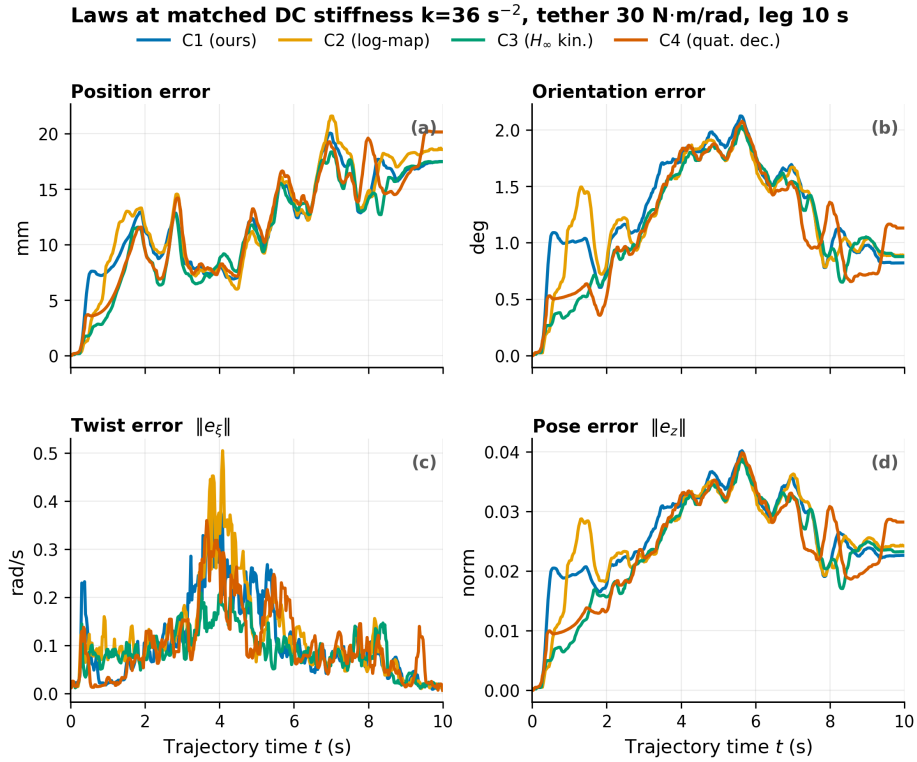


Figure 3: Four laws at equal budget ($k=36$, tether 30 N·m/rad, single runs, 10s nominal trajectory): (a) position error; (b) orientation error; (c) twist error $\|e_\xi\|$; (d) pose error $\|e_z\|$. The four curves coincide in all channels, a hardware confirmation of the equal-information-set equivalence; the contribution of C1 lies in the certificate (Table 1), not in accuracy.

only the proposed law satisfies its attenuation certificate, damping condition, and sublevel margin in all 33 runs. The γ ladder tightens the certified gain from 1/6s to 1/15s (boundary tier 1/18s), with the orientation transient improving monotonically. Meanwhile, the quasi-static residual is set by the balance between the firmware tether spring and friction and does

not vary with the task stiffness—an experimental delineation of the applicability boundary of the static scaling law (5.9).

Limitations and future work: the strictification of the error dynamics (Appendix C.4) is incomplete and the stability conclusions are local; the hardware validation covers a single pose without exogenous disturbance injection, and quantitative on-hardware verification of the static scaling law requires a tether-free setup or a task stiffness well above the tether stiffness; a large-initial-error sweep for the geometric robustness is left as future work.

Appendix C. Supplementary derivations for the control layer

C.1 Explicitness of the disturbance and the role of the α condition

Substituting $\tau = \hat{M}\ddot{\mathbf{q}}_{\text{ref}} + \hat{C}\dot{\mathbf{q}} + \hat{\mathbf{g}}$ into the true dynamics $M\ddot{\mathbf{q}} + C\dot{\mathbf{q}} + \mathbf{g} = \tau + \delta\tau + \tau_{\text{ext}}$ and solving for $\ddot{\mathbf{q}}$:

$$\ddot{\mathbf{q}} = M^{-1}\hat{M}\ddot{\mathbf{q}}_{\text{ref}} + M^{-1}(\Delta C\dot{\mathbf{q}} + \Delta \mathbf{g} + \delta\tau + \tau_{\text{ext}}), \quad M^{-1}\hat{M} = I + M^{-1}\Delta M,$$

which is (5.1). By (5.2), $\ddot{\mathbf{q}}_{\text{ref}}$ is given entirely by the measurable quantities $(\mathbf{q}, \dot{\mathbf{q}}, \hat{\mathbf{x}}_d, \dot{\hat{\mathbf{x}}}_d, \dot{\hat{\mathbf{x}}}_d)$ and does not depend on $\ddot{\mathbf{q}}$; (5.1) is therefore an explicit assignment for $\ddot{\mathbf{q}}$, with no implicit algebraic loop.

The condition $\alpha < 1/\text{cond}_2(K_d)$ in (5.1f) is not a well-posedness condition; it ensures a positive effective damping. From $|e_\xi^\top \Theta(-K_d e_\xi)| \leq \alpha \lambda_{\max}(K_d) \|e_\xi\|^2$ we get $\lambda_{\text{eff}} = \lambda_{\min}(K_d) - \alpha \lambda_{\max}(K_d)$, whose positivity is exactly (5.1f). For larger α the closed loop remains well defined, but the Lyapunov certificate fails.

C.2 Theorem 3(b): singular-value lower bound for A

Step 4 of C.6 requires A to be uniformly invertible on the working set; this is quantified here.

By (4.5), $A = \begin{bmatrix} A_{11} & 0 \\ -[\mathcal{T}]_\times & I_3 \end{bmatrix}$ with $A_{11} = -\frac{1}{2}(\tilde{\eta}I + [\mathcal{O}]_\times)$. Its inverse is

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} & 0 \\ [\mathcal{T}]_\times A_{11}^{-1} & I_3 \end{bmatrix},$$

$$A_{11}^{-1} = -2 \frac{\tilde{\eta}^2 I + \tilde{\eta}[\mathcal{O}]_\times^\top + \mathcal{O}\mathcal{O}^\top}{\tilde{\eta}(\tilde{\eta}^2 + \|\mathcal{O}\|^2)} = -\frac{2}{\tilde{\eta}}(\tilde{\eta}^2 I - \tilde{\eta}[\mathcal{O}]_\times + \mathcal{O}\mathcal{O}^\top),$$

using $\tilde{\eta}^2 + \|\mathcal{O}\|^2 = 1$. Let $B \triangleq \tilde{\eta}I + [\mathcal{O}]_\times$ (so $A_{11} = -\frac{1}{2}B$); then

$$B^\top B = (\tilde{\eta}I - [\mathcal{O}]_\times)(\tilde{\eta}I + [\mathcal{O}]_\times) = \tilde{\eta}^2 I - [\mathcal{O}]_\times^2 = (\tilde{\eta}^2 + \|\mathcal{O}\|^2)I - \mathcal{O}\mathcal{O}^\top = I - \mathcal{O}\mathcal{O}^\top,$$

with eigenvalues $\{\tilde{\eta}^2, 1, 1\}$, so the singular values of B are $\{\tilde{\eta}, 1, 1\}$ and

$$\|A_{11}\|_2 = \frac{1}{2}\sigma_{\max}(B) = \frac{1}{2}, \quad \|A_{11}^{-1}\|_2 = \frac{2}{\sigma_{\min}(B)} = \frac{2}{\tilde{\eta}}.$$

By the block lower-triangular structure, $\|A^{-1}\|_2 \leq (1 + \|\mathcal{T}\|)\|A_{11}^{-1}\|_2 + 1$, hence

$$\sigma_{\min}(A) = \frac{1}{\|A^{-1}\|_2} \geq \left[\frac{2(1 + \|\mathcal{T}\|)}{\tilde{\eta}} + 1 \right]^{-1}.$$

On Ω_c , $\tilde{\eta} \geq \eta_0$ ((5.5c)) and $\|\mathcal{T}\| \leq \sqrt{2c/\lambda_{\min}(K_{p,T})}$, so $\sigma_{\min}(A) \geq c_A(\eta_0, c) > 0$ uniformly, and in step 4 of C.6, $A^\top K_p e_z \equiv 0$ implies $e_z \equiv 0$.

C.3 Details of the split condition in Theorem 3(c-2)

The split rests on two identities: (i) in $(A^\top e_z)_\omega$ the term $[\mathcal{T}]_\times \mathcal{T} = 0$; (ii) in $\dot{\mathcal{T}} = -[\mathcal{T}]_\times \tilde{\omega} + \tilde{v}$ the term $\mathcal{T}^\top (\mathcal{T} \times \tilde{\omega}) = 0$. Both are algebraic identities independent of the working set, so (c-2) is as global as (c-1); if K_d is not block diagonal, $-e_\xi^\top K_d e_\xi$ contains the cross term $\tilde{\omega}^\top K_{\omega v} \tilde{v}$ and the two channel energies no longer separate—one falls back to (c-1).

Necessity of the isotropy of $K_{p,T}$: for a general symmetric positive definite $K_{p,T}$ both identities fail. (i) $(A^\top K_p e_z)_\omega = A_{11}^\top K_{p,O} \mathcal{O} + [\mathcal{T}]_\times K_{p,T} \mathcal{T}$, and $[\mathcal{T}]_\times K_{p,T} \mathcal{T} \neq 0$ unless $\mathcal{T}$ is an eigenvector of $K_{p,T}$; (ii) the derivative of $\frac{1}{2} \mathcal{T}^\top K_{p,T} \mathcal{T}$ contains the coupling term $-\mathcal{T}^\top K_{p,T} [\mathcal{T}]_\times \tilde{\omega}$, which vanishes only for $K_{p,T} = k_{p,T} I_3$. The residual coupling prevents V_ω, V_v from closing individually and one falls back to (c-1).

C.4 Strictification and local ISS (future work)

The gap in Theorem 3(d) stems from the absence of a $-\|e_z\|^2$ term in $\dot{V}$. The standard remedy is strictification: take $W = V + \epsilon e_z^\top K_p A(\tilde{x}) e_\xi$; for small enough ϵ , W is equivalent to V on Ω_c , and its derivative adds $-\epsilon e_z^\top K_p A A^\top K_p e_z$, negative in the e_z direction. If the remaining cross terms (including the $\dot{A}$ term) can be absorbed by the two negative terms, then $\dot{W} \leq -c_1 \|(e_z, e_\xi)\|^2 + c_2 \|d\| \cdot \|(e_z, e_\xi)\|$, i.e., a local ISS-Lyapunov function. This verification (a uniform bound on the $\dot{A}$ term and nonemptiness of the feasible interval for ϵ) is not completed in this paper and is listed among the limitations of §7.

C.6 Complete proof of Theorem 3(b)

1. **Exact cross-term cancellation:** along (5.5) with $d \equiv 0$,

$$\dot{V} = e_\xi^\top \dot{e}_\xi + e_z^\top K_p \dot{e}_z = e_\xi^\top (-K_d e_\xi - A^\top K_p e_z) + e_z^\top K_p A e_\xi = -e_\xi^\top K_d e_\xi,$$

since $e_z^\top K_p A e_\xi = (A^\top K_p e_z)^\top e_\xi$ (only symmetry of K_p is used). Hence $\dot{V} \leq 0$ and Ω_c is forward invariant; $V \leq c$ gives $\|e_\xi\| \leq \sqrt{2c}$ and $\|e_z\| \leq \sqrt{2c/\lambda_{\min}(K_p)}$, so Ω_c is compact.

2. **Sign preservation** (5.5c): $V \leq c$ implies $\frac{1}{2} \mathcal{O}^\top K_{p,O} \mathcal{O} \leq c$, i.e., $\|\mathcal{O}\|^2 \leq 2c/\lambda_{\min}(K_{p,O}) < 1$ (by $c < c^*$). From $\mathcal{O} = -\text{Im } \tilde{r}$ and $\|\tilde{r}\| = 1$ we have $\tilde{\eta}^2 + \|\mathcal{O}\|^2 = 1$, hence $|\tilde{\eta}| \geq \eta_0 > 0$; since $\tilde{\eta}(t)$ is continuous and never zero, $\tilde{\eta}(0) > 0$ gives (5.5c).
3. **Determinant of A :** A is block lower triangular ((4.5)), so

$$\det A = \det\left(-\frac{1}{2}(\tilde{\eta} I_3 + [\mathcal{O}]_\times)\right) \cdot \det I_3 = \left(-\frac{1}{2}\right)^3 \tilde{\eta}(\tilde{\eta}^2 + \|\mathcal{O}\|^2) = -\frac{1}{8} \tilde{\eta},$$

using $\det(aI_3 + [b]_\times) = a(a^2 + \|b\|^2)$ and $\tilde{\eta}^2 + \|\mathcal{O}\|^2 = 1$. The quantitative singular-value bound $\sigma_{\min}(A) \geq [2(1 + \|\mathcal{T}\|)/\tilde{\eta} + 1]^{-1}$ is derived in C.2.

4. **LaSalle:** Ω_c is compact and invariant, and $E \triangleq \{\dot{V} = 0\} \cap \Omega_c = \{e_\xi = 0\} \cap \Omega_c$. If a trajectory stays in E forever: $e_\xi \equiv 0 \Rightarrow \dot{e}_\xi \equiv 0 \Rightarrow A^\top K_p e_z \equiv 0$; by step 3 (A invertible) and $K_p \succ 0$, $e_z \equiv 0$. Hence the largest invariant set in E is $\{(0, 0)\}$, and LaSalle's invariance principle ([Kha02], Thm 4.4) gives (iii).
5. **Local exponential stability:** at $(0, 0)$ we have $\tilde{x} \rightarrow 1$ and $A \rightarrow A_0 = \text{diag}(-\frac{1}{2} I_3, I_3)$; the Jacobian of (5.5) is

$$F = \begin{bmatrix} 0_6 & A_0 \\ -A_0^\top K_p & -K_d \end{bmatrix}.$$

For this LTI system the same V gives $\dot{V} \leq 0$, A_0 is invertible, and repeating step 4 shows F is Hurwitz; the linearization theorem ([Kha02], Thm 4.7) yields local exponential stability. For block-diagonal K_d, K_p , F decouples per channel and the poles are given explicitly by (5.8).

C.7 Complete proof of Theorem 3(c)

Step 1 (exact expression for $\dot{V}$): with disturbances, the cross terms cancel as before, independently of d : $\dot{V} = -e_\xi^\top K_d e_\xi + e_\xi^\top d$ holds exactly.

Step 2 (equivalence in (c-1)): the performance target is $-e_\xi^\top K_d e_\xi + e_\xi^\top d + \frac{1}{2\kappa} \|e_\xi\|^2 - \frac{\gamma_a^2}{2} \|d\|^2 \leq 0$, equivalently the quadratic inequality

$$\begin{bmatrix} e_\xi \\ d \end{bmatrix}^\top M \begin{bmatrix} e_\xi \\ d \end{bmatrix} \geq 0 \quad \forall (e_\xi, d) \in \mathbb{R}^{12},$$

i.e., $M \succeq 0$. Since the lower-right block $\frac{\gamma_a^2}{2} I \succ 0$, the Schur complement gives $K_d - \frac{1}{2\kappa} I - \frac{1}{2\gamma_a^2} I \succeq 0$, which is (5.6a).

Step 3 (global existence): from $M \succeq 0$, $\dot{V} \leq \frac{\gamma_a^2}{2} \|d\|^2$, so $V(t) \leq V(0) + \frac{\gamma_a^2}{2} \|d_{L_2}\|_{L_2}^2 < \infty$; (e_z, e_ξ) is uniformly bounded and the solution exists on $[0, \infty)$ (no finite-time escape).

Step 4 (integration): integrating $\dot{V} \leq -\frac{1}{2\kappa} \|e_\xi\|^2 + \frac{\gamma_a^2}{2} \|d\|^2$ over $[0, T]$, dropping $V(T) \geq 0$, and letting $T \rightarrow \infty$ (monotone convergence) gives (5.6).

Step 5 (per-channel decoupling in (c-2)): by (4.5), $(A^\top K_p e_z)_\omega = A_{11}^\top K_{p,O} \mathcal{O} + k_{p,T} [\mathcal{T}]_\times \mathcal{T} = A_{11}^\top K_{p,O} \mathcal{O}$ (as $\mathcal{T} \times \mathcal{T} = 0$) and $(A^\top K_p e_z)_v = k_{p,T} \mathcal{T}$; moreover, in $\dot{\mathcal{T}} = -[\mathcal{T}]_\times \tilde{\omega} + \tilde{v}$ we have $\mathcal{T}^\top (\mathcal{T} \times \tilde{\omega}) = 0$. Hence the pose cross terms cancel exactly within each channel,

$$\dot{V}_\omega = -\tilde{\omega}^\top K_\omega \tilde{\omega} + \tilde{\omega}^\top d_\omega, \quad \dot{V}_v = -\tilde{v}^\top K_v \tilde{v} + \tilde{v}^\top d_v,$$

and repeating steps 2–4 for each channel ($I_6 \rightarrow I_3$) gives (5.6b) $\Rightarrow$ (5.6').

C.8 Complete proof of Theorem 3(d)

- Expansion of the multiplicative term:** $\dot{V} = -e_\xi^\top K_d e_\xi + e_\xi^\top d$ (step 1 of C.7). Substituting (5.1d) and $u_{fb} = -K_d e_\xi - A^\top K_p e_z$:

$$\dot{V} = -e_\xi^\top K_d e_\xi - \underbrace{e_\xi^\top \Theta K_d e_\xi}_{\text{damping-like}} - \underbrace{e_\xi^\top \Theta A^\top K_p e_z}_{\text{disturbance-like}} + e_\xi^\top d_{ex}.$$

- Separate absorption:** $|e_\xi^\top \Theta K_d e_\xi| \leq \alpha \lambda_{\max}(K_d) \|e_\xi\|^2$ is absorbed into the damping, giving $\lambda_{\text{eff}} > 0$ of (5.7a) by (5.1f); and $|e_\xi^\top \Theta A^\top K_p e_z| \leq \alpha \|A\|_2 \lambda_{\max}(K_p) \|e_z\| \|e_\xi\|$ is absorbed into the equivalent disturbance amplitude D .

- Spectral norm of A :** $A_{11}^\top A_{11} = \frac{1}{4}(I_3 - \mathcal{O}\mathcal{O}^\top)$ gives $\|A_{11}\|_2 = \frac{1}{2}$ (C.2), and the block lower-triangular structure gives $\|A\|_2 \leq \max\{\|A_{11}\|_2, 1\} + \|[\mathcal{T}]_\times\|_2 = 1 + \|\mathcal{T}\|$. On Ω_c , $\|\mathcal{T}\| \leq \sqrt{2c/\lambda_{\min}(K_{p,T})}$ and $\|e_z\| \leq \sqrt{2c/\lambda_{\min}(K_p)}$; substituting into step 2 yields (5.7b) and

$$\dot{V} \leq -\lambda_{\text{eff}} \|e_\xi\|^2 + D \|e_\xi\|. \quad (5.7d)$$

- Young's inequality and integration:** $D \|e_\xi\| \leq \frac{\lambda_{\text{eff}}}{2} \|e_\xi\|^2 + \frac{D^2}{2\lambda_{\text{eff}}}$, so $\dot{V} \leq -\frac{\lambda_{\text{eff}}}{2} \|e_\xi\|^2 + \frac{D^2}{2\lambda_{\text{eff}}}$. Integrating over $[0, T]$ and dropping $V(T) \geq 0$:

$$\frac{\lambda_{\text{eff}}}{2} \int_0^T \|e_\xi\|^2 dt \leq V(0) + \frac{D^2}{2\lambda_{\text{eff}}} T,$$

and dividing by $\frac{\lambda_{\text{eff}}}{2} T$ gives (5.7c); letting $T \rightarrow \infty$ gives (5.7). As $\alpha \rightarrow 0$, $D \rightarrow D_{\text{ex}}$ and $\lambda_{\text{eff}} \rightarrow \lambda_{\min}(K_d)$, recovering the classical form.

C.9 Tightness: the certified gain equals the true linearized gain

Is the certified value of Theorem 3(c') conservative? It is not; the answer is a two-sided squeeze.

The upper bound comes from the certificate (5.6c): any L_2 disturbance leaks into the error at a rate of at most $1/\lambda_{\min}(K_d)$.

The lower bound comes from the frequency response of the linearization. Linearizing the closed loop (5.5) as $\tilde{x} \rightarrow 1$ ((5.8)), the $d \rightarrow e_\xi$ transfer function of each channel is a second-order element $G(s) = \frac{s}{s^2 + bs + c}$: for the rotation channel, $\dot{e}_{z,O} = -\frac{1}{2}\tilde{\omega}$ gives $\tilde{\omega} = -2\dot{O}$, so $b = \lambda(K_\omega)$ and $c = p_O/4$; for the translation channel $\tilde{v} = \dot{T}$, so $b = \lambda(K_v)$ and $c = k_{p,T}$. For G ,

$$|G(j\omega)|^2 = \frac{u}{(c-u)^2 + b^2u}, \quad u = \omega^2, \quad \frac{d}{du} [\cdot] \propto (c-u)(c+u), \quad (5.10)$$

so $|G|$ attains its global maximum $1/b$ at $u = c$, i.e., when the disturbance frequency equals the channel natural frequency $\sqrt{c}$ (independently of the amount of damping). Hence the true H_∞ gain of the linearization with block-diagonal K_d is $\max(1/\lambda(K_\omega), 1/\lambda(K_v)) = 1/\lambda_{\min}(K_d)$.

By the standard result that the local L_2 gain of a nonlinear closed loop is no smaller than the H_∞ norm of its linearization (zero-initial-state, small-signal sense [VdS00]), together with the matching upper bound from the certificate, the local L_2 gain of (5.5) is exactly $1/\lambda_{\min}(K_d)$: the certificate is free of conservatism, and “target $\gamma^* \Leftrightarrow$ damping $\geq 1/\gamma^*$ ” is an exact conversion. A testable by-product: the worst-case disturbance frequency is the channel natural frequency $\sqrt{c}$ (6 rad/s for hardware tier G4).

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